

Investigating the Effects of Synchronized Visuo-Tactile Stimuli for Inducing Kinesthetic Illusion in Observational Learning of Whole-Body Movements

Shintaro Fukumoto

fukumoto.shintaro@irl.sys.es.osaka-u.ac.jp
Osaka University
Toyonaka, Japan

Seiya Mituno

mitsuno.seiya@irl.sys.es.osaka-u.ac.jp
Osaka University
Toyonaka, Japan

Kazuki Nakayama

nakayama.kazuki@irl.sys.es.osaka-u.ac.jp
Osaka University
Toyonaka, Japan

Reiya Itatani

itatani.reiya@irl.sys.es.osaka-u.ac.jp
Osaka University
Toyonaka, Japan

Genki Jogan

jogan.genki@irl.sys.es.osaka-u.ac.jp
Osaka University
Toyonaka, Japan

Mahzoon Hamed

mahzoon@irl.sys.es.osaka-u.ac.jp
Osaka University
Toyonaka, Japan

ABSTRACT

Skill improvement in sports is an essential factor in keeping the motivation to continue. Previous studies showed that kinesthetic illusion enhances observational learning. However, such studies have only dealt with the learning of movements using a single part of the body, and whether kinesthetic illusion can be induced in observational learning of whole-body movements has not been clarified. In this study, we conducted an experiment involving human subjects and confirmed that synchronized visuo-tactile stimuli can induce kinesthetic illusion even in whole-body movements. Moreover, we also demonstrated a complete mediation model in which the synchronization of visuo-tactile stimuli influences kinesthetic illusion mediated by body ownership.

ACM Reference Format:

Shintaro Fukumoto, Seiya Mituno, Kazuki Nakayama, Reiya Itatani, Genki Jogan, and Mahzoon Hamed. 2022. Investigating the Effects of Synchronized Visuo-Tactile Stimuli for Inducing Kinesthetic Illusion in Observational Learning of Whole-Body Movements. In *SIGGRAPH Asia 2022 Posters (SA '22 Posters)*, December 06-09, 2022, ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3550082.3564168>

1 INTRODUCTION

Regular sports activity has significant benefits concerning mental and physical health. Skill improvement in sports is considered important to keep the person motivated to continue. Therefore, adopting a method that helps people to improve their sports performance is expected to lead them to continuously participate in sports activities. In some previous studies, it has been reported that the effects of observational learning are enhanced when the kinesthetic illusion is induced in observers [Nojima et al. 2015]. However, such studies have only dealt with the learning of movements using a single part of the body (e.g. hand, leg, etc.), and it has not been

clarified whether kinesthetic illusion can be induced in observational learning of whole-body movements. Therefore, our research purpose is to investigate the effective methods to induce kinesthetic illusion in observational learning of whole-body movements.

Moreover, Dalila et al [Burin et al. 2017] discussed that body ownership over a fake hand (induced by synchronization of visuo-tactile stimuli) triggered kinesthetic illusion, but whether there is a mediation relationship wasn't statistically verified. Hence, we also aim to clarify the relationship between body ownership and kinesthetic illusion by conducting a mediation analysis to deepen the understanding of how kinesthetic illusion is induced.

2 EXPERIMENT

2.1 Design

We conducted a two-factor within-subjects design experiment that referred to the synchronization of visuo-tactile stimuli with two levels (Sync, Async) and the number/position of visuo-tactile stimuli with three levels (Fig. 2).

2.2 Procedure

A total of 12 male subjects participated in the experiment ($M_{age} = 25.75, SD = 3.22$). The subjects stood upright and observed a male avatar for four minutes. The avatar stood still for the first two minutes and conducted whole-body movements for the last two minutes, which was a baseball swing (Fig. 3). During these periods, subjects were given either synchronous or asynchronous visuo-tactile stimuli. After observing the avatar, the subjects answered a questionnaire about kinesthetic illusion [Nojima et al. 2015] and body ownership [Kalckert and Ehrsson 2012]. This process was repeated for six conditions in random order.

2.3 Result

Fig. 1 illustrates the results of the experiment. First, we conducted a two-way ANOVA and confirmed the main effects of synchronization of visuo-tactile stimuli for both kinesthetic illusion ($F(1, 11) = 6.6, p = .026, \eta^2 = .28$) and body ownership ($F(1, 11) = 17.8, p <$

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

SA '22 Posters, December 06-09, 2022, Daegu, Republic of Korea

© 2022 Copyright held by the owner/author(s).

ACM ISBN 978-1-4503-9462-8/22/12.

<https://doi.org/10.1145/3550082.3564168>

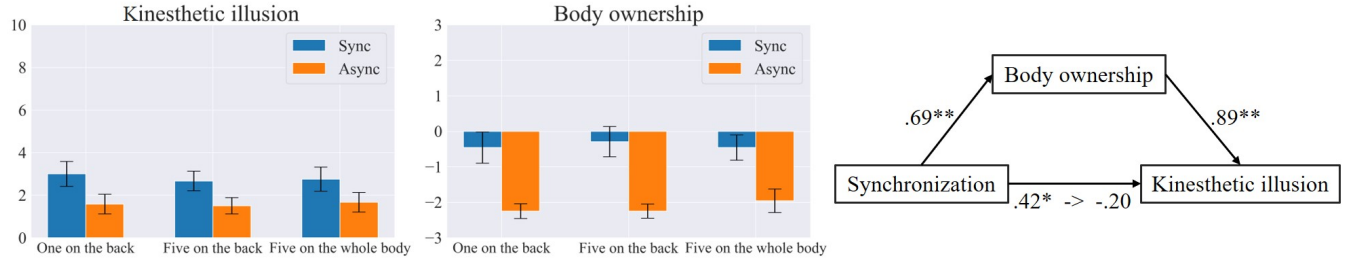


Figure 1: Experimental results (* $p < .05$; ** $p < .01$). Error bars denote standard errors.

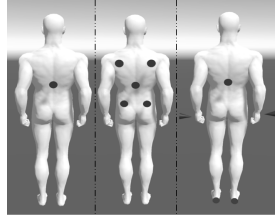


Figure 2: Three types of visuo-tactile stimuli (Left: One on the back, Center: Five on the back, Right: Five on the whole body).

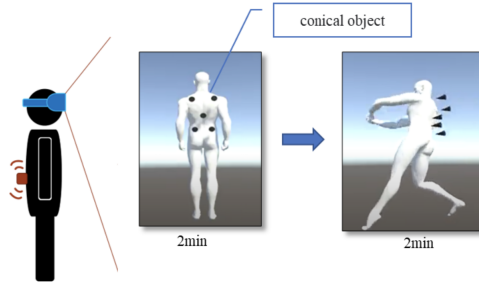


Figure 3: System design which denotes how visuo-tactile stimuli are given. Visual stimuli, in which a conical object touches the avatar, and tactile stimuli, in which a motor vibrates, were used.

.001, $\eta^2 = .62$). The main effect of the number/position of the visuo-tactile stimuli as well as the interaction between the two independent variables were not confirmed neither for the kinesthetic illusion ($F(2, 22) = 0.14, p = .869$ for the main effect, $F(2, 22) = 0.07, p = .929$ for the interaction) nor for the body ownership ($F(2, 22) = 0.09, p = .911$ for the main effect, $F(2, 22) = 0.76, p = .480$ for the interaction).

Next, we conducted a mediation analysis to examine whether synchronization of visuo-tactile stimuli influences the motor illusion mediated by body ownership. Note that, since the main effect of the number/position of visuo-tactile stimuli was not confirmed, the mean scores for this factor were calculated for each of the synchronous and asynchronous conditions. Bootstrapping with 2,000

samples confirmed that the total effect of synchronization of visuo-tactile stimuli on kinesthetic illusion ($\beta = 1.22, b = .42, p = .042$) was reduced to non-significance ($\beta = -0.59, b = -.20, p = .307$) when the indirect effect through body ownership was accounted for ($\beta = 1.82, b = .62, p < .001, 99\%CI [0.63, 3.66]$).

3 CONCLUSION

We confirmed that synchronized visuo-tactile stimuli can induce kinesthetic illusion even in the observational learning of whole-body movements. This result shows that our system may be effective in enhancing observational learning in sports. We also demonstrated a complete mediation model in which the synchronization of visuo-tactile stimuli influences kinesthetic illusion mediated by body ownership, and thus deepened our understanding of the process by which visuo-tactile stimuli induce kinesthetic illusion. In the future, we plan to confirm whether our system can enhance observational learning of whole-body movement through experiments.

ACKNOWLEDGMENTS

This research was supported by “Program for Leading Graduate Schools” of the Osaka University, Japan.

REFERENCES

- Dalila Burin, Maria Pyasik, Adriana Salatino, and Lorenzo Pia. 2017. That’s my hand! Therefore, that’s my willed action: How body ownership acts upon conscious awareness of willed actions. *Cognition* 166 (2017), 164–173. <https://doi.org/10.1016/j.cognition.2017.05.035>
- Andreas Kalckert and Henrik Ehrsson. 2012. Moving a Rubber Hand that Feels Like Your Own: A Dissociation of Ownership and Agency. *Frontiers in Human Neuroscience* 6 (2012), 40. <https://doi.org/10.3389/fnhum.2012.00040>
- Ipppei Nojima, Satoko Koganemaru, Toshio Kawamata, Hidenao Fukuyama, and Tatsuya Mima. 2015. Action observation with kinesthetic illusion can produce human motor plasticity. *European Journal of Neuroscience* 41, 12 (2015), 1614–1623.